Status	Finished
Started	Thursday, 29 January 2026, 7:42 PM
Completed	Thursday, 29 January 2026, 8:07 PM
Duration	24 mins 43 secs
Grade	**44.00** out of 97.00 (**45.36**%)

Question **1**

Correct

Mark 15.00 out of 15.00

1. Linked list is an example of Dynamic type of memory allocation.

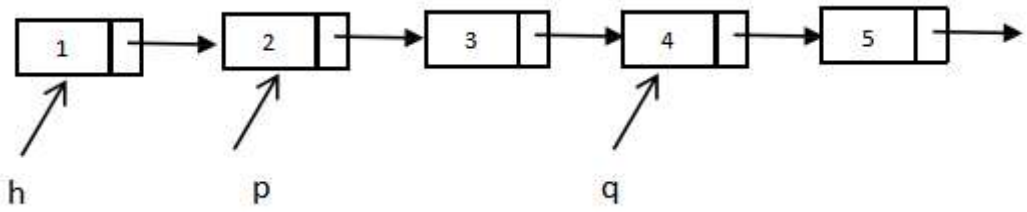
2. In a pointer based linked list, each node contain minimum of 2 fields, which are to store the actual data , and a pointer to the next node , no answer here .

3. Which of the following is the best way to implement the Link List to satisfy the request: "Let me access the item at position n"?

Singly linked list (Array based implementation)

4. Given the folowing linked list, find the output when the code executes.

```
struct node {
    int data;
    node* next; };
```




```
int main(){
  cout << p->next->next->next>data;
  cout << h->next->next->data;
  cout << p->data;
  cout << q->data; }//end main
```

Answer :

5 3 2 4

5. Suppose that p and q point to adjacent nodes (p before q) in a linked list of nodes defined by

class nodeType. What is the missing statement so that a new node is inserted between p and q.

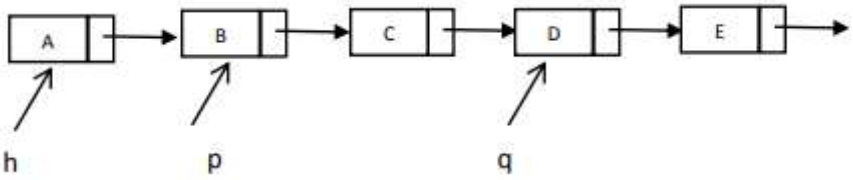
```
class nodeType {
  public:
    int info;
    nodeType *next; };
  nodeType *head, *p, *q, *newNode;
  newNode = new nodeType;
```

newNode->next=q;

```
  p->next = newNode;
```

6. In folowing linked list what is the output of portion code?

```
struct node {
  int data;
  node* next; };
```




```
cout << h->next->next->next>data
cout << p->next->data
cout << q->data
cout << p->next->next->data
```

Answer:

D C D D

Your answer is correct.

The correct answer is:

1. Linked list is an example of [Dynamic] type of memory allocation.

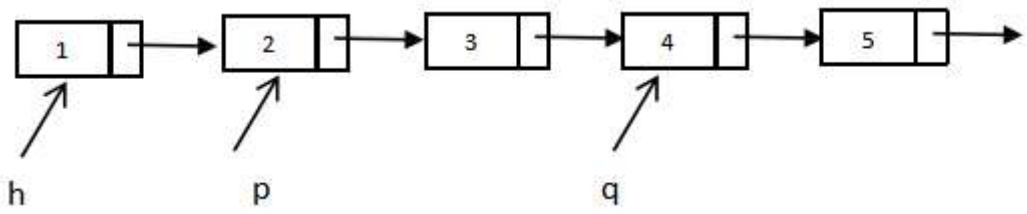
2. In a pointer based linked list, each node contain minimum of [2] fields, which are to store the [actual data] , [and a pointer to the next node] , [no answer here].

3. Which of the following is the best way to implement the Link List to satisfy the request: "Let me access the item at position n"?

[Singly linked list (Array based implementation)]

4. Given the folowing linked list, find the output when the code executes.

```
struct node {
    int data;
    node* next; };
```




```
int main(){
  cout << p->next->next->next>data;
  cout << h->next->next->data;
  cout << p->data;
  cout << q->data; }//end main
```

Answer :

[5] [3] [2] [4]

5. Suppose that p and q point to adjacent nodes (p before q) in a linked list of nodes defined by

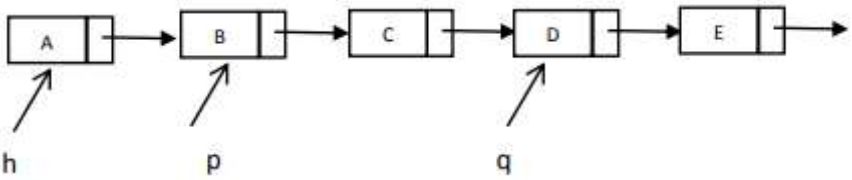
class nodeType. What is the missing statement so that a new node is inserted between p and q.

```
class nodeType {
```

```
public:
  int info;
  nodeType *next; };
nodeType *head, *p, *q, *newNode;
newNode = new nodeType;
[newNode->next=q;]
p->next = newNode;
```

6. In folowing linked list what is the output of portion code?

```
struct node {
  int data;
  node* next; };
```




```
cout << h->next->next->next>data
cout << p->next->data
cout << q->data
cout << p->next->next->data
```

Answer:

[D][C][D][D]

Question **2**

Partially correct

Mark 15.00 out of 29.00

Evaluate the following postfix expression using stack concept

3 9 % 20 * 7 -

Postfix	Ch	Operator	Operand1	Operand 2	Result	Stack
3 9 % 20 * 7 -	NONE	NONE	NONE	NONE	NONE	NONE
9 % 20 * 7 -	3 ✓					3 ✓
% 20 * 7 -	9 ✓					3, 9 ✗ , 3 ✗
20 * 7 -	% ✓	% ✓	3 ✗	9 ✗	3 ✓	3, 20 ✗
* 7 -	20 ✓					60 ✗ , 60, 7 ✗
7 -	* ✓	* ✓	3 ✗	20 ✗	60 ✓	53 ✗
-	7 ✓					53 ✗ , 53 ✗
	- ✓	- ✓	60 ✗	7 ✗	53 ✓	53 ✓

Question 3

Partially correct

Mark 14.00 out of 27.00

Using stack operations, show step by step conversion of the following infix expression to postfix notation.

A – B * (C + D * E) / F

NOTE :

1. Write NONE if the Stack or Postfix Notation is empty.

2. All answers (character) including NONE must be written continuously without any whitespace and comma. For example : ABC+-

Infix Notation	Stack	Postfix Notation
A – B * (C + D * E) / F	#NONE	NONE
A – B * (C + D * E) / F	# NONE ✓	NODE NONE NONE ✗
– B * (C + D * E) / F	# NONE ✗	A ✓
B * (C + D * E) / F	# - ✓	A ✗
* (C + D * E) / F	# - ✓	AB ✓
(C + D * E) / F	# -* ✗	AB ✓
C + D * E) / F	# -*(✓	AB ✗
+ D * E) / F	# -*(✗	ABC ✓
D * E) / F	# -*(+ ✓	ABC ✗
* E) / F	# -*(+ ✓	ABCD ✓
E) / F	# -*(+* ✓	ABCD ✗
) / F	# -*(+* ✓	ABCDE ✗

/ F	# -*	ABCDE*+
F	#	

Question **4**

Not answered

Marked out of 26.00

a) Given a search function as below. [5 marks]

```
int search( int search_key, int array[ ],
int array size) _size)
{ int p;
int index = -1; //-1 means record not found
for (p = 0; p < array_size; p++)
{ if ( _ y y [p] search key < array [p] )
break;
// loop terminats when the value of
// search key is smaller than the current array
// element
else if(search key== array[p]) _key== array[p]) // key is found
{ index = p;
// assign current index
break;
} // end else-if
}//end for
return index;
// return the value of index
} //e
```

Based on the given function, fill in the blanks with appropriate value when the function is executed.

p = ⊗ index = ⊗ array[p] = ⊗

search_key = 23

[0]	[1]	[2]	[3]	[4]
11	22	32	33	45

p = ⊗ index = ⊗ array[p] = ⊗

search_key = 23

[0]	[1]	[2]	[3]	[4]
11	22	32	33	45

p = ⊗ index = ⊗ array[p] = ⊗ return = ⊗

search_key = 23

[0]	[1]	[2]	[3]	[4]
11	22	32	33	45

b) What is the time complexity (Big O notation) the above function? O ⊗ , [2 marks]

c) Given a search function as below. [6 marks]

```
int search( int searchkey, int array_size, const int array [] )
{   bool found = false;
int index = -1;
int MIDDLE, LEFT = 0, RIGHT = array_size-1;
while (( LEFT<= RIGHT ) && (!found))
{ MIDDLE = (LEFT + RIGHT )/ 2;
if ( array[MIDDLE] == search_key)
{index = MIDDLE; found = true;
} else if (array[MIDDLE] > search_key)
RIGHT= MIDDLE- 1;
else LEFT= MIDDLE+ 1;
} //end while
return index;
}//end function
```

Based on the given function, fill in the blanks with appropriate value when the function is executed.

LEFT = ⊗ MIDDLE = ⊗ RIGHT = ⊗

array[LEFT] = ⊗

search_key = 12

[0]	[1]	[2]	[3]	[4]
2	3	5	9	12

LEFT = ⊗ MIDDLE = ⊗ RIGHT = 4

array[LEFT] = 9

search_key = 12

[0]	[1]	[2]	[3]	[4]
2	3	5	9	12

LEFT = 4 MIDDLE = ⊗ RIGHT = 4 array[LEFT] = 12

return = ⊗

search_key = 12

[0]	[1]	[2]	[3]	[4]
2	3	5	9	12

d) What is the time complexity (Big O notation) the above function? O

⊗ , [2 marks]

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sample tree